Assignments 1(Database Creation and Insertion of Records)

- **1) Create a Database with a name Customer_Details in SQL ?**

Assignments 1

- **2) Create a Customer table which comprises of these columns with appropriate data type :**
  - **customer_id**
  - **first_name**
  - **last_name**
  - **email**
  - **address**
  - **city**
  - **state**
  - **zip**

Assignments 1

- **3) Insert 5 new records into the Customer table based on data type**
- **4) Select only the first_name & last_name columns from the Customer table.**
- **5) Select only those records where first_name starts with “G” and city starts with ‘Sa’.**

Assignment 2(Joins)

- 1) Create an Orders table which comprises of these columns :
  - order_id
  - order_date
  - Amount
  - customer_id
- 2) Make an inner join on Customer & Order tables on the customer_id column.

Assignment 2

- **3) Make left and right joins on ‘Customer’ & Order tables on the ‘customer_id’ column.**
- **4) Update the Orders table, set the amount to be 100 where customer_id is 3.**

Assignment 3(SQL Aggregate Commands)

- **1) Use the inbuilt functions and find the minimum, maximum and average amount from the orders table..**
- **2) Create a user-defined function, which will multiply the given number with 10.**
- **3) Use the case statement to check if 100 is less than 200, greater than 200 or equal to 200 and print the corresponding value**

Assignment 4(SQL Selection Commands: Ordering & Filtering)

- 1) Arrange the Orders dataset in decreasing order of amount.
- 2) Create a table with name Employee_details1 and comprising of these columns :
  - Emp_id
  - Emp_name
  - Emp_salary

Assignment 4

- **3) Create another table with name Employee_details2, which comprises of same columns as first table using (script table as option with new query editor window).**
- **4) Apply the union operator, intersect operator, except operator on these two tables.**

Assignment 5 (SQL Views)

- 1) Create a view named Customer_san_jose which comprises of only those customers who are from San Jose City.
- 2) Inside a table, need to update the first name of the customer to Francis, where the last name is Jordan.
- 3) Set the first name of customer to Alex, where the last name is Jordan
- 4) Inside a try catch block, divide 100 with 0, print the default error message

Assignment 5

- **Link**
- **https://inblog.in/SQL-Assignments-MuHnRIPq3i**

SQL Date & Time Assignment

- **1) Write a query to display the first day of the month (in datetime format) three months before the current month?**
  - **Sample current date : 2014-09-03**

    Expected result : 2014-06-01
- **2) Write a query to display the last day of the month (in datetime format) three months before the current month?**

SQL Date & Time Assignment

- **3) Write a query to get the distinct Mondays from hire_date in employees tables?**
  - **Sample Table : Employees**

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT
100	Steven	King	SKING	515.123.4567	1987-06-17	AD_PRES	24000.00	0.00
101	Neena	Kochhar	NKOCHHAR	515.123.4568	1987-06-18	AD_VP	17000.00	0.00
102	Lex	De Haan	LDEHAAN	515.123.4569	1987-06-19	AD_VP	17000.00	0.00
103	Alexander	Hunold	AHUNOLD	590.423.4567	1987-06-20	IT_PROG	9000.00	0.00
104	Bruce	Ernst	BERNST	590.423.4568	1987-06-21	IT_PROG	6000.00	0.00
105	David	Austin	DAUSTIN	590.423.4569	1987-06-22	IT_PROG	4800.00	0.00
106	Valli	Pataballa	VPATABAL	590.423.4560	1987-06-23	IT_PROG	4800.00	0.00

SQL Date & Time Assignment

- 4) Write a query to get the first day of the current year?
- 5) Write a query to get the last day of the current year?
- 6) Write a query to calculate the age in year?
- 7) Write a query to get the current date in the following format?
  - Sample date : 2014-09-04

    Output : September 4, 2014

SQL Date & Time Assignment

- 8) Write a query to get the current date in Thursday September 2014 format?
  - Thursday September 2014
- 9) Write a query to extract the year from the current date?
- 10) Write a query to get the DATE value from a given day (number in N)?
  - Sample days : 730677

    Output : 2000-07-11

SQL Date & Time Assignment

- **11) Write a query to get the first name and hire date from employees table where hire date between '1987-06-01' and '1987-07-30‘?**
  - **Sample Table : Employees**

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT
100	Steven	King	SKING	515.123.4567	1987-06-17	AD_PRES	24000.00	0.00
101	Neena	Kochhar	NKOCHHAR	515.123.4568	1987-06-18	AD_VP	17000.00	0.00
102	Lex	De Haan	LDEHAAN	515.123.4569	1987-06-19	AD_VP	17000.00	0.00
103	Alexander	Hunold	AHUNOLD	590.423.4567	1987-06-20	IT_PROG	9000.00	0.00
104	Bruce	Ernst	BERNST	590.423.4568	1987-06-21	IT_PROG	6000.00	0.00
105	David	Austin	DAUSTIN	590.423.4569	1987-06-22	IT_PROG	4800.00	0.00
106	Valli	Pataballa	VPATABAL	590.423.4560	1987-06-23	IT_PROG	4800.00	0.00

SQL Date & Time Assignment

- 12) Write a query to display the current date in the following format?
  - Sample output: Thursday 4th September 2014 00:00:00
- 13) Write a query to display the current date in the following format?
  - Sample output: 05/09/2014
- 14) Write a query to display the current date in the following format?
  - Sample output: 12:00 AM Sep 5, 2014

SQL Date & Time Assignment

- **15) Write a query to get the firstname, lastname who joined in the month of June?**
  - **Sample Table : Employees**

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT
100	Steven	King	SKING	515.123.4567	1987-06-17	AD_PRES	24000.00	0.00
101	Neena	Kochhar	NKOCHHAR	515.123.4568	1987-06-18	AD_VP	17000.00	0.00
102	Lex	De Haan	LDEHAAN	515.123.4569	1987-06-19	AD_VP	17000.00	0.00
103	Alexander	Hunold	AHUNOLD	590.423.4567	1987-06-20	IT_PROG	9000.00	0.00
104	Bruce	Ernst	BERNST	590.423.4568	1987-06-21	IT_PROG	6000.00	0.00
105	David	Austin	DAUSTIN	590.423.4569	1987-06-22	IT_PROG	4800.00	0.00
106	Valli	Pataballa	VPATABAL	590.423.4560	1987-06-23	IT_PROG	4800.00	0.00

SQL Date & Time Assignment

- **16) Write a query to get the years in which more than 10 employees joined?**
  - **Sample Table : Employees**

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT
100	Steven	King	SKING	515.123.4567	1987-06-17	AD_PRES	24000.00	0.00
101	Neena	Kochhar	NKOCHHAR	515.123.4568	1987-06-18	AD_VP	17000.00	0.00
102	Lex	De Haan	LDEHAAN	515.123.4569	1987-06-19	AD_VP	17000.00	0.00
103	Alexander	Hunold	AHUNOLD	590.423.4567	1987-06-20	IT_PROG	9000.00	0.00
104	Bruce	Ernst	BERNST	590.423.4568	1987-06-21	IT_PROG	6000.00	0.00
105	David	Austin	DAUSTIN	590.423.4569	1987-06-22	IT_PROG	4800.00	0.00
106	Valli	Pataballa	VPATABAL	590.423.4560	1987-06-23	IT_PROG	4800.00	0.00

SQL Date & Time Assignment

- **17) Write a query to get first name of employees who joined in 1987?**
  - **Sample Table : Employees**

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT
100	Steven	King	SKING	515.123.4567	1987-06-17	AD_PRES	24000.00	0.00
101	Neena	Kochhar	NKOCHHAR	515.123.4568	1987-06-18	AD_VP	17000.00	0.00
102	Lex	De Haan	LDEHAAN	515.123.4569	1987-06-19	AD_VP	17000.00	0.00
103	Alexander	Hunold	AHUNOLD	590.423.4567	1987-06-20	IT_PROG	9000.00	0.00
104	Bruce	Ernst	BERNST	590.423.4568	1987-06-21	IT_PROG	6000.00	0.00
105	David	Austin	DAUSTIN	590.423.4569	1987-06-22	IT_PROG	4800.00	0.00
106	Valli	Pataballa	VPATABAL	590.423.4560	1987-06-23	IT_PROG	4800.00	0.00

SQL Date & Time Assignment

- **18) Write a query to get department name, manager name, and salary of the manager for all managers whose experience is more than 5 years?**
  - **Sample Table : Departments**

DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
10	Administration	200	1700
20	Marketing	201	1800
30	Purchasing	114	1700
40	Human Resources	203	2400
50	Shipping	121	1500
60	IT	103	1400
70	Public Relations	204	2700
80	Sales	145	2500

SQL Date & Time Assignment

- **19) Write a query to get employee ID, last name, and date of first salary of the employees?**
  - **Sample Table : Employees**

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT
100	Steven	King	SKING	515.123.4567	1987-06-17	AD_PRES	24000.00	0.00
101	Neena	Kochhar	NKOCHHAR	515.123.4568	1987-06-18	AD_VP	17000.00	0.00
102	Lex	De Haan	LDEHAAN	515.123.4569	1987-06-19	AD_VP	17000.00	0.00
103	Alexander	Hunold	AHUNOLD	590.423.4567	1987-06-20	IT_PROG	9000.00	0.00
104	Bruce	Ernst	BERNST	590.423.4568	1987-06-21	IT_PROG	6000.00	0.00
105	David	Austin	DAUSTIN	590.423.4569	1987-06-22	IT_PROG	4800.00	0.00
106	Valli	Pataballa	VPATABAL	590.423.4560	1987-06-23	IT_PROG	4800.00	0.00

MANAGER_ID	DEPARTMENT_ID
0	90
100	90
100	90
102	60
103	60
103	60
103	60

SQL Date & Time Assignment

- **20) Write a query to get first name, hire date and experience of the employees?**
  - **Sample Table : Employees**

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT
100	Steven	King	SKING	515.123.4567	1987-06-17	AD_PRES	24000.00	0.00
101	Neena	Kochhar	NKOCHHAR	515.123.4568	1987-06-18	AD_VP	17000.00	0.00
102	Lex	De Haan	LDEHAAN	515.123.4569	1987-06-19	AD_VP	17000.00	0.00
103	Alexander	Hunold	AHUNOLD	590.423.4567	1987-06-20	IT_PROG	9000.00	0.00
104	Bruce	Ernst	BERNST	590.423.4568	1987-06-21	IT_PROG	6000.00	0.00
105	David	Austin	DAUSTIN	590.423.4569	1987-06-22	IT_PROG	4800.00	0.00
106	Valli	Pataballa	VPATABAL	590.423.4560	1987-06-23	IT_PROG	4800.00	0.00

MANAGER_ID	DEPARTMENT_ID
0	90
100	90
100	90
102	60
103	60
103	60
103	60

SQL Date & Time Assignment

- **21) Write a query to get the department ID, year, and number of employees joined?**
  - **Sample Table : Employees**

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT
100	Steven	King	SKING	515.123.4567	1987-06-17	AD_PRES	24000.00	0.00
101	Neena	Kochhar	NKOCHHAR	515.123.4568	1987-06-18	AD_VP	17000.00	0.00
102	Lex	De Haan	LDEHAAN	515.123.4569	1987-06-19	AD_VP	17000.00	0.00
103	Alexander	Hunold	AHUNOLD	590.423.4567	1987-06-20	IT_PROG	9000.00	0.00
104	Bruce	Ernst	BERNST	590.423.4568	1987-06-21	IT_PROG	6000.00	0.00
105	David	Austin	DAUSTIN	590.423.4569	1987-06-22	IT_PROG	4800.00	0.00
106	Valli	Pataballa	VPATABAL	590.423.4560	1987-06-23	IT_PROG	4800.00	0.00

MANAGER_ID	DEPARTMENT_ID
0	90
100	90
100	90
102	60
103	60
103	60
103	60

SQL Answers

- 1) SELECT date(((PERIOD_ADD (EXTRACT(YEAR_MONTH FROM CURDATE()),-3)*100)+1));
- 2) SELECT (SUBDATE(ADDDATE (CURDATE(),INTERVAL 1 MONTH), INTERVAL DAYOFMONTH(CURDATE())DAY)) AS LastDayOfTheMonth;
- 3) SELECT DISTINCT(STR_TO_DATE (CONCAT(YEARWEEK(hire_date),'1'),'%x%v%w')) FROM employees;

SQL Date & Time Assignment Answers

- 4) SELECT MAKEDATE(EXTRACT(YEAR FROM CURDATE()),1);

- 5) SELECT STR_TO_DATE(CONCAT(12,31, EXTRACT(YEAR FROM CURDATE())), '%m%d%Y');

- 6) SELECT YEAR(CURRENT_TIMESTAMP) - YEAR("1967-06-08") - (RIGHT(CURRENT_TIMESTAMP, 5) < RIGHT("1967-06-08", 5)) as age;

SQL Date & Time Assignment Answers

- 7) SELECT DATE_FORMAT(CURDATE(),'%M %e, %Y') AS 'Current_date';
- 8) SELECT DATE_FORMAT(NOW(), '%W %M %Y');
- 9) SELECT EXTRACT(YEAR FROM NOW());
- 10) SELECT FROM_DAYS(730677);

SQL Date & Time Assignment Answers

- **11) SELECT FIRST_NAME, HIRE_DATE FROM employees WHERE HIRE_DATE BETWEEN '1987-06-01 00:00:00' AND '1987-07-30 23:59:59';**

- **12) SELECT date_format(CURDATE(),'%W %D %M %Y %T');**

- **13) SELECT date_format(CURDATE(),'%d/%m/%Y');**

- **14) SELECT date_format(CURDATE(),'%l:%i %p %b %e, %Y');**

SQL Date & Time Assignment Answers

- **15) SELECT first_name, last_name FROM employees WHERE MONTH(HIRE_DATE) = 6;**

- **16) SELECT DATE_FORMAT(HIRE_DATE,'%Y') FROM employees GROUP BY DATE_FORMAT(HIRE_DATE,'%Y') HAVING COUNT(EMPLOYEE_ID) > 10;**

- **17) SELECT FIRST_NAME, HIRE_DATE FROM employees WHERE YEAR(HIRE_DATE)=1987;**

SQL Date & Time Assignment Answers

- **18) SELECT DEPARTMENT_NAME, FIRST_NAME, SALARY FROM departments D JOIN employees E ON (D.MANAGER_ID=E.MANAGER_ID) WHERE (SYSDATE()-HIRE_DATE) / 365 > 5;**

- **19) SELECT employee_id, last_name, hire_date, LAST_DAY(hire_date) FROM employees;**

SQL Date & Time Assignment Answers

- **20) SELECT FIRST_NAME, SYSDATE(), HIRE_DATE, DATEDIFF(SYSDATE(), hire_date)/365 FROM employees;**

- **21) SELECT DEPARTMENT_ID, DATE_FORMAT(HIRE_DATE,'%Y'), COUNT(EMPLOYEE_ID) FROM employees GROUP BY DEPARTMENT_ID, DATE_FORMAT(HIRE_DATE, '%Y') ORDER BY DEPARTMENT_ID;**